

Design and development of automated mobile robotic arm vehicle for cold storage operations

1st Mr. Omkar Pore
dept.of Mechanical engineering
VPKBIET, Baramati
Baramati, India
omkarpore03@gmail.com

2nd Mr. Girish Patil
dept.of Mechanical Engineering
VPKBIET, Baramati
Baramati, India
girishpatil1705@gmail.com

3rd Mr. Utkarsh Jadhav
dept.of Mechanical Engineering
VPKBIET, Baramati
Baramati, India
utkarshjadhav994@gmail.com

4th Mr. Subodh Bhosale
dept.of Mechanical Engineering
VPKBIET, Baramati
Baramati, India
subodhbhosale0803@gmail.com

5th prof. Suraj Kumbhar
dept.of Mechanical Engineering
VPKBIET, Baramati
Baramati, India
surajkumbhar6@gmail.com

Abstract— The development of an automatic mobile robotic arm for cold storage operations is aimed at enhancing efficiency, accuracy and security in the environment, where temperature-sensitive items are handled. This robotic system is designed to autonomously navigate cold storage features, pick, sort and store items and operate tasks such as storing and operating in tight temperature conditions. This innovation is particularly important for industries such as food and pharmaceuticals, where it is important to maintain the integrity of temperature-sensitive products. Robotic arm dynamics allow it to make it favourable to various cold storage configurations, while its exact control mechanisms ensure safe handling of delicate objects. The system not only strengthens operations, but also improves security by reducing human risk for extremely cold environment, which leads to better overall productivity and cost savings.

Keywords— Automatic mobile robotic arm, robotic system

I. INTRODUCTION

Cold storage warehouses play a crucial role in supply chains for perishable goods such as food, pharmaceuticals, and chemicals. However, extreme cold temperatures (as low as -30°C) make human labor inefficient, causing fatigue, reduced productivity, and health risks. Traditional robotic solutions are often fixed in place, limiting adaptability in dynamic storage environments. This research aims to develop a mobile robotic arm system capable of navigating cold storage environments, identifying goods, and performing automated retrieval and placement operations. The system is designed to work autonomously or via remote control, using AI-based object recognition, LiDAR-based navigation, and real-time environmental adaptation.

II. LITERATURE SURVEY

A. literature review

- Cold Storage Challenges in Warehouse Automation.

Cold storage environments present unique challenges for automation, primarily due to extreme temperatures affecting mechanical components, electronics, and human labor. Studies by Bakker et al. (2021) highlight that worker fatigue and health risks significantly impact operational efficiency in sub-zero environments. Traditional robotic systems struggle with condensation, low battery performance, and material brittleness at extremely low temperatures (Kim & Lee, 2020).

- Robotic Systems for Warehouse Automation

The application of robotics in warehouses has been widely explored, particularly through Automated Storage and

Retrieval Systems (AS/RS). Research by Zhong et al. (2019) emphasizes that AS/RS improves efficiency and reduces labor costs, but these systems are typically fixed, limiting adaptability in dynamic environments. Autonomous Mobile Robots (AMRs), such as those developed by Amazon Robotics and Fetch Robotics, have shown promise in general warehouse automation, but studies indicate they require modifications for cold storage, including insulation and specialized sensors (Gao et al., 2022).

- Mobile Robotic Arms in Logistics

Mobile robotic arms combine the benefits of mobility and precise object manipulation. Wurman et al. (2017) explored the use of robotic arms for logistics applications, noting improvements in order fulfillment speed. However, their effectiveness in cold storage has not been extensively studied. More recent advancements, such as those by Gupta et al. (2023), demonstrate AI-powered mobile robots performing autonomous retrieval and placement, but without adaptations for sub-zero conditions.

- AI-Based Object Recognition for Cold Environments

AI-based vision systems have been widely adopted in warehouse automation, with Convolutional Neural Networks (CNNs) and deep learning models proving effective in object detection and classification (Redmon & Farhadi, 2018). However, cold storage conditions introduce challenges such as frost accumulation, low lighting, and visual obstructions due to fogging (Chen et al., 2021). Solutions such as infrared imaging and sensor fusion have been proposed to enhance object recognition reliability in extreme conditions (Huang et al., 2022).

- LiDAR-Based Navigation in Cold Storage

LiDAR-based navigation enables precise mapping and localization in warehouse environments. Research by Thrun et al. (2020) demonstrates that LiDAR combined with SLAM (Simultaneous Localization and Mapping) enhances autonomous robot navigation. However, cold storage warehouses with narrow aisles and reflective surfaces introduce mapping inaccuracies (Park et al., 2023). Newer approaches, such as multi-sensor fusion combining LiDAR with thermal and ultrasonic sensors, have been suggested to improve navigation robustness in low-temperature environments.

- Real-Time Environmental Adaptation in Robotics

Real-time environmental adaptation is crucial for autonomous operations in dynamic storage environments. Khatib (2022) explored adaptive control algorithms for robotic manipulators, emphasizing their role in adjusting to temperature-induced mechanical variations. Machine learning-driven adaptation strategies have been proposed by Liu et al. (2023), where reinforcement learning enables robots to modify movement patterns based on environmental feedback, reducing operational failures in extreme conditions. The development of an Automated Mobile Robotic Arm for Cold Storage Operations follows a structured and systematic approach to ensure efficient material handling, precise object recognition, and real-time adaptation to harsh environmental conditions. The process involves a series of research, design, prototyping, testing, and optimization phases to create a reliable and cost-effective robotic solution tailored for cold storage warehouse automation.

III. PROBLEM IDENTIFICATION

Cold storage warehouses require efficient automation due to extreme environmental conditions that affect both human labour and traditional robotic systems. While mobile robotic arms offer flexibility in material handling, their deployment in sub-zero environments presents several critical challenges

A. Environmental Challenges in Cold Storage

Extreme Temperatures: Mechanical components become brittle, lubricants lose efficiency, and batteries Experience reduced performance in freezing conditions. **Condensation & Frost Formation:** Rapid temperature fluctuations lead to moisture build up, affecting sensors, Electronics, and vision systems. **Low Visibility & Lighting Issues:** Cold storage warehouses often have dim lighting, And fogging further obstructs cameras and LiDAR sensors.

B. Limitations of Current Robotics in Cold Storage

- **Fixed AS/RS Systems:** Traditional Automated Storage and Retrieval Systems (AS/RS) lack adaptability in dynamic warehouse settings.
- **AMR Challenges:** Autonomous Mobile Robots (AMRs) used in standard warehouses require modifications, such as insulation and specialized sensors, to function effectively in cold environments. **Lack of Mobile Manipulation:** Existing robotic arms used in logistics are often stationary and not optimized for mobility in cold storage.

C. Key Development Phases:

- **Research & Requirements Analysis:**

Identify challenges related to low temperatures, condensation, and limited human intervention in cold storage environments. Study AI-based object recognition, LiDAR-based navigation, and reinforcement learning for real-time adaptation. Analyze material compatibility for cold-resistant components, sensors, and actuators to ensure longevity.

- **Design & System Architecture:**

Develop a mobile robotic platform with an integrated robotic arm capable of operating in sub-zero conditions. Design an AI-driven vision system for recognizing, localizing, and manipulating objects accurately. Incorporate insulation and heating elements to prevent sensor and mechanical failures due to extreme cold.

- **Prototyping & Hardware Integration:**

Build a functional prototype with industrial-grade components suited for cold environments. Integrate LiDAR, depth cameras, and AI-based vision for precise navigation and object interaction. Develop a real-time control system with reinforcement learning-based adaptation for dynamic warehouse conditions.

Testing & Performance Evaluation:

Conduct extensive cold storage trials to assess system robustness, energy efficiency, and automation reliability. Optimize gripping mechanisms to ensure stable handling of frozen goods and temperature-sensitive items. Evaluate battery performance and power management under extreme cold conditions.

- **Optimization & Deployment:**

Enhance AI models for improved object detection and navigation accuracy. Refine motion planning algorithms for real-time adjustments to dynamic warehouse layouts. Implement seamless integration with existing warehouse management systems (WMS) for autonomous operation.

IV. MODELING AND WORKING

A. Design Explanation:

1. Structural Design

Material: The arm is made from 3D-printed plastic (likely PLA or ABS), making it lightweight yet sturdy.

Segmented Arm: The design includes multiple articulated sections, mimicking industrial robotic arms.

2. Actuation & Motion

Servo Motors: Several hobby-grade servo motors (visible in black casings) control the different joints, enabling movement in multiple degrees of freedom. **Gear System:** The base houses a visible orange gear mechanism that helps in rotating the arm.

3. Mechanical Linkages

Parallel Linkage Mechanism: The arm utilizes parallel linkages to ensure smooth and controlled movement of the end effector. **Bolted Joints:** Screws and metal bolts reinforce key pivot points for added durability.

4. End Effector (Gripper)

Servo-Powered Claw: The gripper is controlled by a small servo motor, enabling it to grasp objects.

5. Base and Mounting

Sturdy Base: The base is designed to be mounted or placed on a flat surface for stability.

Mounting Holes: Visible at the bottom for securing the arm to a surface.

B. Conceptual Design & Prototyping

The conceptual design and layout is developed using SolidWorks. This phase involved creating a comprehensive 3D model that integrated key components such as sensors, extinguishing mechanisms, and housing structures. By leveraging SolidWorks, we refined the spatial arrangement of each component to achieve a compact and Efficient layout that aligns with the structural framework of an electric vehicle. This design process facilitated a Thorough analysis of spatial constraints, ultimately enhancing both the

functionality and safety of the fire Suppression system. After immense prototyping & improvements the finalization of design is done & the design Includes all the major key components of the system.

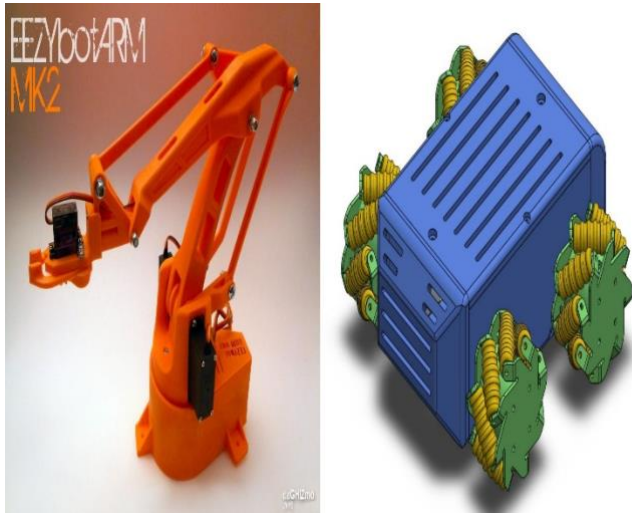


Fig. Conceptual design of robot.

C. Layout & Its Key Features

Layout

The following design is made, and the image gives an idea about the layout of all the components.



Fig. Assembly of robotic system

Key features

1. Integrated Mobile Base & Robotic Arm – Combines an articulated robotic arm with an omnidirectional mobile Platform.
2. Omnidirectional Mecanum Wheels – Enables smooth and precise movement in any direction for efficient Warehouse navigation.
3. Compact & Modular Design – Ensures easy integration into automated logistics and warehouse operations.
4. Servo-Driven Arm with Multi-DOF – Provides precise object manipulation and picking capabilities.

5. AI-Driven Navigation & Object Recognition – Uses LiDAR and vision sensors for real-time mapping and object Detection.
6. Cold-Resistant Construction – Designed to operate in low-temperature environments with protected Electronics.
7. Versatile End-Effector – Capable of handling various object types with adaptive gripping mechanisms.
8. Autonomous Operation – Can be programmed for self-guided tasks, reducing manual intervention.

Working

The mobile robotic arm operates seamlessly with a mobile controller system, enabling remote and autonomous Control for warehouse automation. The controller interfaces via Wi-Fi, Bluetooth, or a dedicated RF module, Allowing real-time communication between the user and the robot. Using a custom mobile app or joystick-based Interface, operators can manually control movement, adjust arm positioning, and activate object-gripping Functions. The system also supports autonomous mode, where AI-driven navigation and object recognition guide Operations based on pre-programmed tasks. Equipped with LiDAR and sensor fusion, the robotic arm adapts to Obstacles dynamically. The mobile controller enhances usability, flexibility, and efficiency, making the system ideal for cold storage and logistics automation

Pick and Place Process:

1. The pick and place operation of the mobile robotic arm follows a structured sequence, integrating AI-driven Perception, precise actuation, and autonomous navigation for efficient object handling.
2. Object Detection & Localization – Using AI-based vision and LiDAR sensors, the system identifies and locates Objects within its workspace.
3. Navigation & Positioning – The Mecanum wheel base moves the robot to the optimal position, ensuring precise Alignment with the target object.
3. Arm Actuation & Gripping – The multi-DOF robotic arm extends toward the Object, and the adaptive gripper (vacuum, soft, or electromagnetic) securely grasps it.
4. Transport & Repositioning – The robotic system moves to the designated drop-off location, optimizing its path Through autonomous route planning.
5. Object Placement & Release – The arm accurately positions and releases the object at the specified location Before returning to its idle state or next task.

Analysis & Calculations :

1. Payload Capacity and Overload protection :

This is the maximum weight that robotic arm is designed to lift (assumption) and safety margin has to 1.3 more than load.

$$P_{\text{safe}} = P_{\text{exact load}} \times \text{safety factor.}$$

$$P_{\text{safe}} = 200 \text{ mg} \times 1.3 = 130 \text{ mg} = 260 \text{ mg}$$

2. Torque Calculation for Arm Joints :

$$T = F \times r \dots \dots \dots (F = m \times g = 0.200 \times 9.81 = 1.962 \text{ N})$$

$$T = 1.962 \times 50 = 98.1 \text{ N.cm} \dots \dots (r = 50 \text{ cm overall height})$$

$$T = 10.00 \text{ kg.cm}$$

3. Motor Power :

$$P = T \times \omega / 1000$$

$$P = 0.981 \times (2\pi/t) / 1000 \dots \dots \dots (t = 3 \text{ sec})$$

$$P = 0.00205 \text{ Kw} \dots \dots \dots (P = 2.05W)$$

Result :

For the development of an automated mobile robotic arm in cold storage operations:

- The payload capacity should be designed for at least 0.26 kg with an overload protection factor.
- The joint torque requirement is 10.00 kg.cm (98.1 N.cm) for smooth operation.
- The motor power requirement is 2.05 W, ensuring energy-efficient operation.

These calculations provide a fundamental basis for selecting appropriate actuators, ensuring reliability in cold storage environments. Further refinements should consider frictional losses, joint dynamics, and environmental adaptations.

Conclusion :

The integration of an automated mobile robotic arm, such as the EezyBotArm MK2, with an omnidirectional mobile platform presents a promising solution for enhancing efficiency in industrial and specialized environments like cold storage warehouses. This combination enables precise object manipulation and flexible mobility, allowing seamless navigation in constrained spaces. The robotic system's modularity and adaptability make it suitable for various applications, from logistics to precision assembly. Future improvements could focus on AI-driven automation, advanced sensor integration, and IoT connectivity for real-time monitoring and optimization. This development contributes to the ongoing advancement of smart robotics, improving productivity and operational safety.

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